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# **DATABASE**

## **MODULE 0 - *LEARNING ROADMAP & COURSE OBJECTIVES***

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## INTRODUCTION

### OBJECTIVES

1. Basic Database Concepts: Learn the basic concepts that form the foundation of database systems.
2. Relational Data and Non-Relational Data: Understanding the differences between relational and non-relational data.
3. Topics in the Database Practicum Module: Understanding each topic in the database module.
4. Software installation: installing the software needed for the database practicum.

### MODULE TARGETS

1. Understanding the Learning Structure: Practitioners will study each module (there are 6 main modules) for one week.

### PREPARATION

1. Muslim students may recite the following prayer before starting the lesson  
 رَضِيتُ بِاللّهِ رَبًّا وَبِالْإِسْلَامِ دِينًا وَبِمُحَمَّدٍ نَبِيًّا وَرَسُولًا. رَبِّ زِدْنِي عِلْمًا وَارْزُقْنِي فَهْمًا
2. Laptop or personal computer
3. **Oracle SQL Developer Data Modeler** installed and ready to use.  
[LINK-SQL\\_Developer\\_Data\\_Modeler](#)  
**Oracle 11g XE:**  
[LINK\\_ORACLE\\_11G\\_XE](#)  
**Oracle APEX:**  
[LINK\\_ORACLE\\_APEX](#)
4. Strong motivation and commitment to learning

### KEYWORDS

CDM, PDM, DDL, DML, Normalization Concepts, Function, Join, Subquery, View, Oracle 11g XE



## INTRODUCTION TO DATABASE MATERIAL

### 1. What is a database?



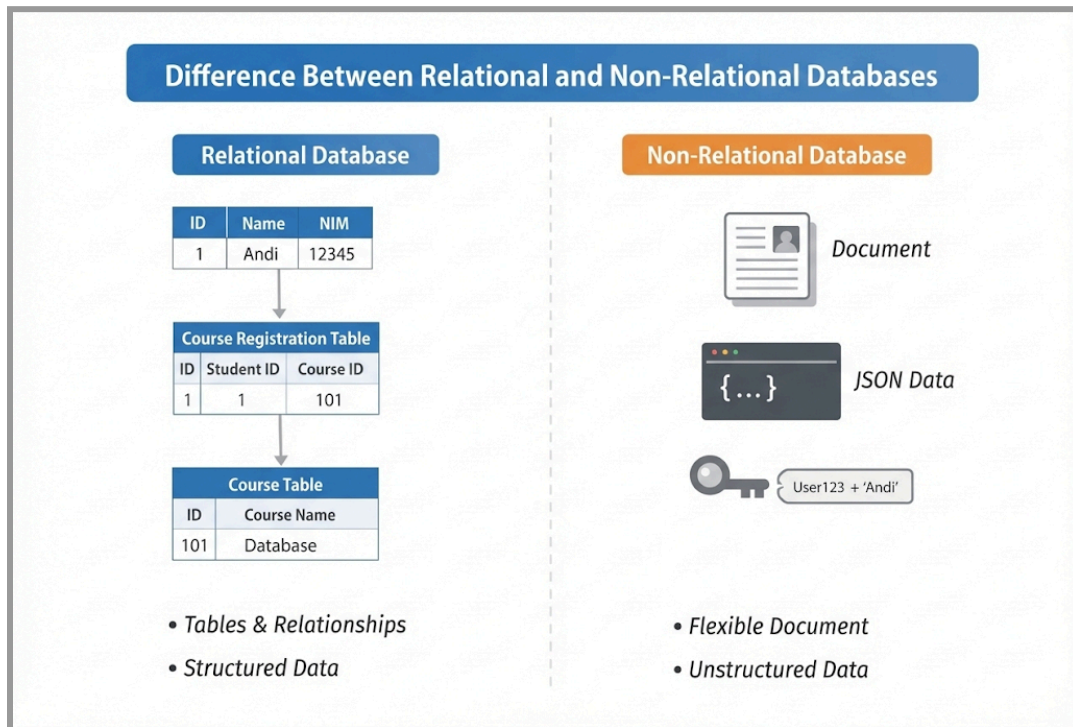
A **database** is a collection of data stored in a structured manner in a computer system and managed using a **Database Management System (DBMS)**, so that data can be stored, accessed, updated, and managed efficiently. Compared to conventional data storage such as text files or spreadsheets, databases are capable of handling large amounts of data, reducing duplication, and maintaining data consistency and security. Databases are widely used in various information systems, such as academic systems, banking, e-commerce, hospitals, and web-based and mobile applications.

### 2. Why Should You Learn Databases?

Databases are an important component in almost all modern information systems. By studying databases, students can understand how to **store, manage, and process data in a structured and efficient** manner, so that data can be used optimally to support decision making. Database knowledge also forms the basis for application development, both web-based and desktop-based, as almost all applications require consistent, secure, and easily accessible data management. Therefore, understanding databases is essential for building reliable and integrated information systems.



### 3. Differences Between Relational and Non-Relational Databases



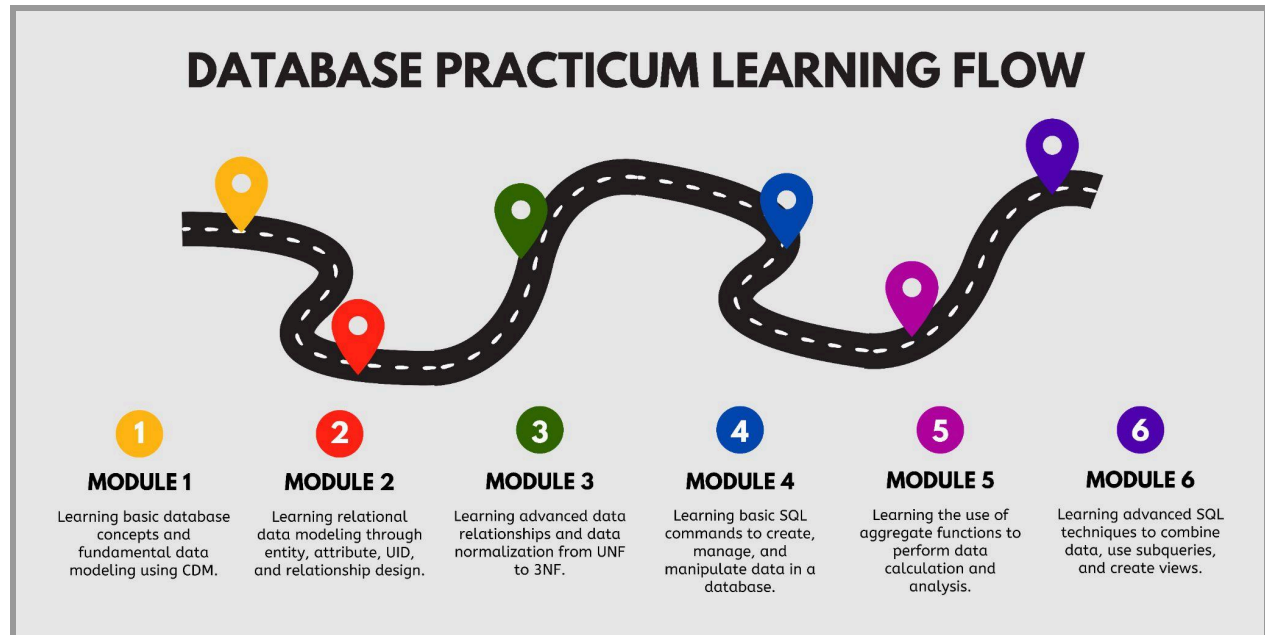
**Relational databases** store data in the form of **tables** that are interconnected with one another. Each table has rows and columns, and uses keys to link data between tables. This type of database is suitable for **structured and organized data**, such as student data, grades, and transactions.

**Non-relational databases** store data with a **more flexible** structure that is not always in table form. Data can be stored in various formats, such as documents or JSON, making it suitable for data that is constantly changing and does not always have the same structure, such as chat data, comments, or user activity.

**In short**, relational databases emphasize order and relationships between data, while non-relational databases emphasize flexibility in data storage. In this practicum, learning is focused on relational databases.



## PRACTICUM LEARNING FLOW



### MODULE 1 - BASIC CONCEPT OF DATABASE

- Explain the basic concepts of data and databases.
- Explain the components of a database system.
- Explain the purpose and benefits of using databases.
- Explain the basic concepts of data modeling.
- Explain entities, attributes, and attribute characteristics.
- Explain the concept of Conceptual Data Model (CDM).
- Provide examples of simple CDM design using Oracle Data Modeler.

### MODULE 2 - RELATIONAL DATA MODELING

- Explain the concept of relational data modeling.
- Explain entities, attributes, and Unique Identifiers (UID).
- Explain relationships between entities (1:1, 1:N, and M:N).
- Explain cardinality and optionality in relationships.
- Explain the use of data types in database modeling.
- Explain the application of precision and scale for numeric attributes.
- Provide examples of relational CDM design using Oracle Data Modeler.



### MODULE 3 - RELATIONAL AND DATA NORMALIZATION

- Explain advanced data relationships (supertype–subtype, ARC, recursive, transferable, and barred relationships).
- Explain the process of converting CDM into table structures.
- Explain the concept and purpose of data normalization.
- Explain normalization stages from UNF to 3NF.
- Explain the differences between tables at each normalization stage.
- Explain the process of converting normalized tables back into CDM.
- Provide examples of normalization and CDM comparison before and after normalization.

### MODULE 4 - BASIC SQL COMMANDS

- Explain the basic concepts of **Structured Query Language (SQL)** and its functions in database management.
- Explain **Data Definition Language (DDL)** commands such as **CREATE**, **ALTER**, and **DROP**.
- Explain **Data Manipulation Language (DML)** commands such as **INSERT**, **UPDATE**, and **DELETE**.
- Explain the use of basic **SELECT** commands to display data from tables.
- Explain the use of **WHERE** as a data filter based on specific conditions.
- Explain the correct and systematic structure of writing SQL queries.
- Provide examples of the use of basic SQL commands in Oracle database 11g XE.

### MODULE 5 - AGGREGATE FUNCTION

- Explaining the concept of **Aggregate Functions** and their use in data processing.
- Explaining the types of **Aggregate Functions** such as **COUNT**, **SUM**, **AVG**, **MIN**, and **MAX**.
- Explaining the use of Aggregate Functions in **SELECT** commands.
- Explaining the use of the **GROUP BY** clause for data grouping.
- Explaining the use of the **HAVING** clause for filtering aggregation results.
- Explaining the concept of sorting data using the **ORDER BY** clause.
- Providing examples of the use of Aggregate Functions in Oracle database 11g XE.

### MODULE 6 - JOIN, SUBQUERY AND VIEW

- Explaining the use of **JOIN** to combine data from two or more tables.
- Explaining the types of **JOIN** such as **INNER JOIN**, **LEFT JOIN**, and **RIGHT JOIN**.
- Explaining the concept and function of **Subquery** in SQL queries.
- Explaining the use of **Subquery** in **SELECT**, **WHERE**, and **FROM** commands.
- Explaining the concept of **Views** as virtual tables.
- Explaining the benefits of using **Views** for data simplification and security.
- Providing examples of the use of **JOIN**, **SUBQUERIES** AND **VIEWS** in Oracle database 11g XE.



## SOFTWARE INSTALLATION

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**Please download and install the Oracle software on your own device.**

**Oracle SQL Developer Data Modeler 24.3.1 (Used for modules 1 to 3)**

[LINK-SQL\\_Developer\\_Data\\_Modeler](#)

**Oracle 11g XE (Used for modules 4 to 6)**

[LINK\\_ORACLE 11G XE](#)

For users of **macOS** or other operating systems who cannot use Oracle 11g XE, it is recommended to use Oracle APEX as an alternative. Oracle APEX has similar functions to Oracle 11g XE and is used in Modules 4 to 6 as a medium for practicing writing and executing SQL queries. In addition, **Oracle APEX will also be used in the Final Practicum Exam (UAP)**. Therefore, the use of Oracle APEX is highly recommended for practicum participants to familiarize themselves with the exam environment.

**Oracle APEX**

[LINK\\_ORACLE APEX](#)

